TRINETRA

MCD Unified Cyber Defense System

Problem Statement & Architectural Diagram — Detailed Report

SECTION 1 | PROBLEM STATEMENT

The Problem TRINETRA Solves

> ***"Rightnow,afakeMCDwebsiteisliveontheinternetcollectingpayments from Delhi citizens. MCD does not know it exists."***

1.1 The Context — Why This Problem Exists Now

Municipal Corporation of Delhi (MCD) is the largest urban local body in India, serving over 2 crore citizens across Delhi. It manages critical citizen services digitally — property tax payment, building permits, contractor payments, grievance portals, health licenses, and more. Every day, lakhs of Delhi residents interact with MCD's digital infrastructure.

This digital expansion has made MCD a high-value target. Attackers know that government portals carry real financial data, personal information, and Aadhaar-linked records. They also know that municipal corporations historically have the weakest cybersecurity posture of any government body — large enough to be a valuable target, under-resourced enough to be an easy one.

1.2 The Scale of the Problem — Validated Numbers

22.68 Lakh	₹36,450 Cr	21 Days	138%
Cyber Incidents India 2024	Fraud Losses Reported 2024	Avg Fake Govt Site Stays Live	Rise in Govt Cyberattacks (2019–23)

These are not projections. These are reported figures from NCRP, CERT-In, and The Cyber Express covering 2024 and 2025. The problem is accelerating, not stabilizing.

1.3 The Two Realities MCD Lives In Simultaneously

MCD operates two digital realities every single day — and currently has zero visibility into either one.

The Outside Threat	The Inside Threat	The Real Gap
• Fake MCD websites collecting citizen payments in real time • Phishing campaigns on WhatsApp and SMS reaching lakhs of Delhi residents • Average fake government site stays LIVE for 21 days undetected • MCD has ZERO systematic process to find these sites	• 2,400 computers across 12 zones — no unified security view • 40 IT staff managing everything with zero automated monitoring • Login attack in South Zone and East Zone on same morning = same attacker — nobody connects them • Attacks go undetected for hours or days	• These two threats are CONNECTED — not separate problems • Fake site harvests citizen passwords at 9am • Real MCD portal attacked at 10am using those same passwords • No existing tool in India sees both sides simultaneously

1.4 Reality 1 — What Is Happening Outside MCD Right Now

On the public internet, at this moment, there are websites designed to look exactly like MCD's official portals. These sites are built using pixel-perfect copies of MCD's design — same colors, same logos, same layout, same forms. They are promoted through Google ads targeting searches like "MCD property tax payment Delhi." They receive real traffic from real Delhi residents who believe they are on the official government website.

- A citizen visits the fake site, enters their username and password, and submits a payment.
- The attacker captures those credentials instantly.
- The citizen receives either an error page or a fake success message.
- MCD receives no notification. The citizen reports it weeks later, if at all.
- The fake site continues operating. Average duration before detection — 21 days.

MCD has no systematic process to find these sites. The current method is citizen complaints — which arrive after the damage is done.

1.5 Reality 2 — What Is Happening Inside MCD's Network Right Now

Inside MCD's network, 2,400 computers across 12 geographically separated zones generate thousands of security-relevant events every single day — login attempts, file accesses, network connections, privilege changes. This data exists. It is being generated automatically by Windows Event Logs and web server logs on every MCD server.

- A login anomaly in South Zone and a credential stuffing attack in East Zone on the same morning may be the same attacker probing multiple entry points simultaneously.
- Currently, nobody connects these events because nobody is watching both zones simultaneously through a single unified system.
- 40 IT staff manage all 12 zones with no automated threat correlation across zones.
- Attacks that span multiple zones are invisible to any single zone's administrator.

1.6 The Connection Nobody Is Seeing — The Core Gap

> **External and internal attacks are not separate events.They are phases of the same attack.**
>
> 9:00AM—Attacker's fake MCD property tax website goes live.Citizens begin entering their credentials.
>
> 9:45AM—Attacker harvests 300 username-password pairs from the fake site.
>
> 10:00AM—Automated credentials tuffing attack begins onthereal MCD portal using the harvested passwords.
>
> **No existing tool connects the set events.TRINETRAdoes.**

1.7 Why Existing Solutions Fail

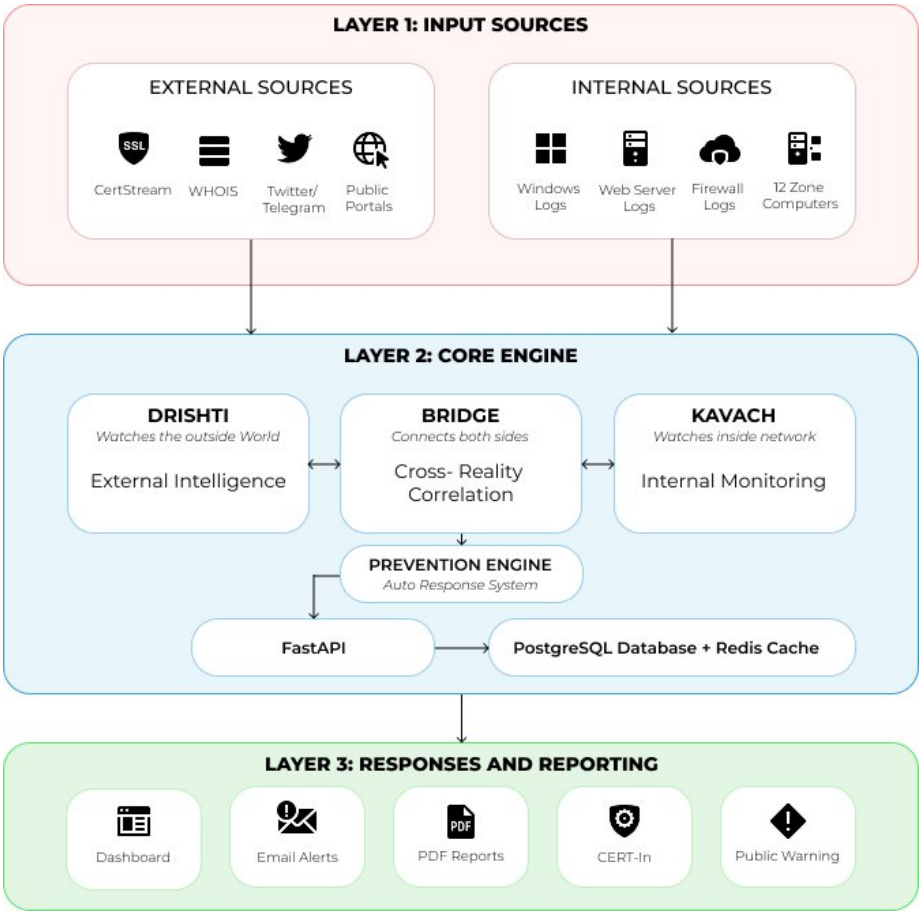
Solution	What It Does	Critical Gap
CloudSEK	Brand & fake site monitoring	Only watches externally. No internal network visibility. Cannot connect external fake site to internal attack.
TAC Security / Lucideus	Internal vulnerability assessment	One-time audits, not real-time. Cannot detect live external impersonation campaigns.
Wazuh (SIEM)	Internal log aggregation	Blind to the public internet. Cannot detect fake sites or phishing campaigns targeting citizens.
CERT-In	Government cyber response	Reactive only — acts after complaints arrive. No proactive detection. Average response time measured in days.

1.8 The One-Sentence Problem Definition

> ***MCD has zero visibility into fake websites impersonating it on the public internet AND zero unified visibility into attacks happening inside its own network — and no tool today connects these two threats even though attackers use them as phases of a single coordinated campaign.***

Architecture Explained — Layer by Layer

The TRINETRA architecture is organized into three clear layers that flow from top to bottom — Input Sources, Core Engine, and Outputs. Each layer has a single, clear responsibility. Together they create a capability that no existing tool has.



2.1 The Three Core Modules

DRISHTI	THE BRIDGE	KAVACH
External Intelligence	*Cross-Reality Correlation*	*Internal Monitoring*
Watches the entire public internet for threats against MCD. Runs without any access to internal systems.	Connects DRISHTI and KAVACH. Finds the relationship between external and internal events automatically.	Watches MCD's internal network across all 12 zones. Reads existing logs — nothing new installed.

2.2 Layer 1 — Input Sources (The Eyes of TRINETRA)

This layer answers one question: where does TRINETRA's data come from? It has two sides — external and internal — running simultaneously and independently. Neither side requires the other to function.

External Sources (Left Side of Layer 1)

- CertStream — Every SSL certificate registered anywhere in the world appears in a public real-time data stream within minutes of registration. TRINETRA monitors this stream 24 hours a day, 7 days a week. When a new domain containing MCD-related keywords registers an SSL certificate, TRINETRA catches it within minutes — often before the attacker has even finished setting up the fake site.
- WHOIS — When a suspicious domain is found, TRINETRA queries its registration details. A government portal domain registered 4 minutes ago at 11pm on a Tuesday is inherently suspicious. Domain age is a strong signal included in the confidence score.
- Twitter and Telegram — TRINETRA monitors public social media for MCD-related keywords combined with fraud indicators. When 50 accounts share the same suspicious link within 2 hours, that is a coordinated phishing campaign — not random noise.
- Public Portals — TRINETRA probes all MCD public-facing portal URLs every 5 minutes from the outside, the same way a citizen's browser would. It measures response time and error rates against a 30-day baseline. When a DDoS attack begins, TRINETRA knows before the first citizen complaint arrives.

Internal Sources (Right Side of Layer 1)

- Windows Event Logs — Every MCD server running Windows automatically generates detailed security audit logs — every login attempt, every failed authentication, every privilege change. These logs already exist. TRINETRA reads them without creating any new data.
- Web Server Logs — Apache and IIS logs from MCD's public portals record every incoming request. A credential stuffing attack generates a distinctive pattern — hundreds of requests per minute, each with a different username, automated timing. TRINETRA detects this pattern from the logs.
- Firewall Logs — Where available, firewall logs show connection attempts blocked at the network perimeter. These supplement Windows logs to provide a complete picture of network activity.
- 12 Zone Computers — Delhi's MCD is divided into 12 administrative zones. Each zone has one lightweight Python agent installed on a single server. This agent reads local logs and forwards structured alert records to the central hub over HTTPS. Installation takes 2 to 3 hours per zone.

2.3 Layer 2 — Core Engine (The Brain of TRINETRA)

This is where data becomes intelligence. Three modules process their respective inputs and one central bridge connects them. Everything runs inside a FastAPI hub that coordinates all components through a PostgreSQL database and Redis cache.

DRISHTI — External Intelligence Module

DRISHTI processes everything coming from the external sources. It runs a four-stage pipeline on every suspicious domain:

- Stage 1 — Domain Similarity Scoring using Levenshtein Distance. Every candidate domain is mathematically compared against all 19 legitimate MCD domains. A domain that requires only 1 character edit to match a real MCD domain is almost certainly a typosquat.
- Stage 2 — Visual Comparison using Perceptual Hashing (dHash). A headless browser takes a screenshot of the suspicious site. The screenshot is converted to a 64-bit mathematical fingerprint and compared against stored fingerprints of real MCD portals. Two screenshots that differ by only 4 bits out of 64 are 93.75% visually identical.
- Stage 3 — HTML Content Analysis using BeautifulSoup. The page's source code is parsed to find Aadhaar input fields, payment forms, MCD-specific text patterns, and mismatched logo sources.
- Stage 4 — Composite Scoring. Domain similarity (30%), visual similarity (35%), content analysis (25%), and domain age (10%) combine into a single confidence score. Above 75% is confirmed fake. 50–75% is probable. Below 50% is monitored only.

KAVACH — Internal Monitoring Module

KAVACH processes everything coming from inside MCD's network. It applies five deterministic detection rules to log data from all 12 zones:

- Rule 1 — Failed Login Spike: More than 50 failed logins within any 10-minute window. Automated credential stuffing scripts generate exactly this pattern.
- Rule 2 — Off-Hours Privileged Access: Any administrator account login between 10pm and 6am. Legitimate administrative work happens during business hours.
- Rule 3 — Port Scan Signature: A single IP address attempting connections to more than 20 different ports within 5 minutes. This is textbook network reconnaissance.
- Rule 4 — Large Outbound Data Transfer: Any outbound transfer exceeding 500MB within one hour. MCD servers have no legitimate reason to send hundreds of megabytes externally in a short window.
- Rule 5 — Foreign Server Connection: Any MCD server initiating a connection to an IP address geolocated outside India using MaxMind GeoIP2. MCD servers have no legitimate reason to contact servers in Eastern Europe or Southeast Asia.

KAVACH also runs a Cross-Zone Correlation Engine every 15 minutes. If the same alert type appears in more than two geographically separated zones within 2 hours, it generates a Coordinated Attack Alert — escalating from three individual yellow alerts to one red alert that explicitly identifies a single attacker targeting multiple zones simultaneously.

THE BRIDGE — Cross-Reality Correlation Engine (The Genuine USP)

The Bridge is what makes TRINETRA unique. It runs every 10 minutes and asks one question: has DRISHTI detected a confirmed fake site targeting a specific portal type in the last 4 hours, AND has KAVACH detected a login spike on the real version of that same portal type in the same window?

- If both conditions are true — SAME portal type, SAME time window — The Bridge generates a Cross-Reality Alert. This is the highest priority alert class in the system.
- The Canary Credential System adds forensic certainty. When a fake site is confirmed, TRINETRA automatically submits a unique trackable credential to the fake site's login form. If that exact credential later appears in login attempt logs on the real portal, the connection is forensically proven — not inferred. This is court-admissible evidence.
- Three confidence tiers exist — Probable (timing only), Likely (timing plus burst pattern), and Confirmed Forensic (canary credential triggered). Each tier triggers a different response protocol.

FastAPI Hub, PostgreSQL, and Redis

FastAPI is the central nervous system that coordinates all components. It exposes ingestion endpoints that receive data from all zone collectors and DRISHTI pipelines, runs scheduled jobs for The Bridge correlation, cross-zone correlation, and weekly reports, and serves the React dashboard.

PostgreSQL stores six core tables: fake_sites, alerts, campaigns, portal_health, cross_reality_events, and canary_credentials. Every alert from every source is stored here. The Bridge's SQL query joins fake_sites and alerts tables with a time window filter.

Redis serves two purposes: publishing real-time alert events to all open dashboard browsers via Server-Sent Events with under 2 seconds latency, and serving as the Celery task broker for computationally intensive background operations such as screenshot capture and PDF generation.

Prevention Engine — Auto Response System

When The Bridge fires a Cross-Reality Alert, the Prevention Engine automatically generates response assets without waiting for human input — IP block lists formatted for direct firewall copy-paste, takedown request PDFs pre-addressed to CERT-In and the domain registrar, and public warning notices drafted for MCD's official social media channels. A human officer reviews and executes these outputs. The knowledge is automated. The authority remains human.

2.4 Layer 3 — Responses and Reporting (The Voice of TRINETRA)

This layer answers one question: once TRINETRA has detected something, how does it communicate to the people who need to act? Five output channels ensure the right information reaches the right person at the right time.

Dashboard	Email Alerts	PDF Reports	CERT-In Package	Public Warning
Live zone map, real-time alert feed, cross-reality timeline	Instant notification to IT Head on HIGH or CRITICAL events	Auto-generated weekly report every Monday 6am to Commissioner	Pre-filled evidence bundle, one-click government submission	Citizen alert draft, one-click publish on MCD official channels

2.5 The Complete Data Flow — One Attack from Start to Finish

11:23 PM	Attacker registers fake MCD domain. SSL certificate issued.
11:27 PM	CertStream delivers certificate to TRINETRA. Keyword match fires in under 1 microsecond.
11:34 PM	PhantomWatch completes 4-stage analysis. Confidence 84%. CONFIRMED FAKE. Dashboard alert fires. Canary credential submitted automatically.
12:47 AM	Attacker begins credential stuffing on real MCD portal using 340 harvested passwords. KAVACH Rule 1 fires — 600 failed logins in 3 minutes.

12:50 AM	The Bridge runs correlation check. Finds fake site from 11:34 PM and login spike from 12:47 AM. Same portal type. 73-minute gap. Cross-Reality Alert — CONFIDENCE HIGH.
12:51 AM	Canary credential appears in real portal login logs. The Bridge upgrades alert to FORENSIC CONFIRMED. Evidence package finalized.
1:00 AM	IT Officer follows 5 numbered steps in dashboard. IPs blocked. Rate limiting enabled. Takedown filed. Public warning published. Total response time — 23 minutes.

2.6 Why This Architecture Wins

- No permission required to start — DRISHTI monitors the public internet from day one. MCD does not need to install, approve, or deploy anything for TRINETRA to begin delivering value.
- Explainable intelligence — every alert contains a plain English explanation of what happened, why it is suspicious, and exactly what to do next in numbered steps. No security expertise required to act on TRINETRA alerts.
- No black-box AI — every detection model is mathematically explainable. A CAG auditor can verify every score. A judge can understand every alert. Levenshtein distance, dHash similarity, Z-score anomaly — all produce numbers with traceable sources.
- Forensic-grade evidence — the canary credential system produces court-admissible proof linking a specific fake site to a specific internal attack. This moves TRINETRA from a monitoring tool to a law enforcement enablement platform.
- Scales linearly — adding a new MCD zone is 3 hours of installation work and zero architectural changes. Adding a new city is a configuration change. The platform handles 500 zone collectors without redesign.

TRINETRA is the only system that watches MCD from both sides simultaneously — and automatically connects external threats to internal attacks when they are phases of the same coordinated campaign. That capability does not exist anywhere else.